



World's First All-Weather Autonomous Bus

A new autonomous all-weather bus designed to drive even in heavy rain, snow, or fog is currently rolling out in Finland. <https://digit.in/apr19-15>



Levitating objects with light

Researchers levitate, propel objects using only light, by creating nanoscale patterning on the objects' surfaces. <https://digit.in/apr19-16>

periods of our cosmos. At an altitude of 36 kilometers, the atmosphere is tenuous enough for high altitude balloons to study the cosmos, without picking up any interference. This was the target altitude for ARCADE to study the heat emissions from the oldest stars in the Universe. The scientists observing the readings from the instruments came across something unexpected – radio waves six times more intense than what was expected. The “expectation” here is the total amount of radio energy detected from all known sources in the universe. The space roar phenomenon has been described both as being louder, and brighter in scientific literature, but refer to the same thing. Detailed analysis of the signal ruled out known sources. The radio waves were not from early stars, from any known radio source, or from the halo of the Milky Way – a bubble of thin gasses and widely scattered stars. Heat from the instruments on ARCADE itself were also ruled out as a source for the noise. The farther away we peer into the universe, the farther back in time we see, so whatever is causing the space roar, is tied in to the earliest times. The space roar prevents astronomers from studying the earliest stars, as the radio waves are the only way to study them, and there is a lot of noise. Future missions can be calibrated to ignore the space roar, allowing scientists to study the signals from the earliest stars.

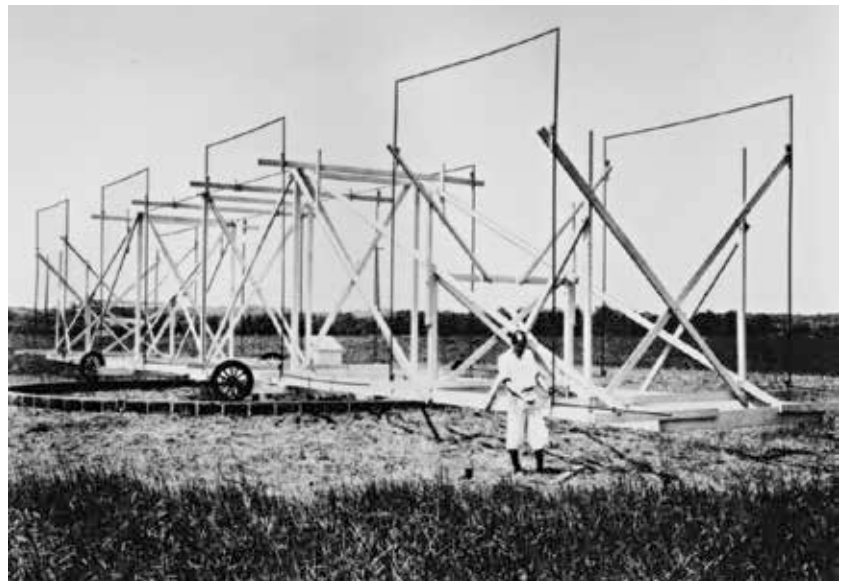
Alan Kogut, the leader of the team that sent up the balloon noted, “instead of the faint signal we hoped to find, here was this booming noise six times louder than anyone had predicted.” The amount of radio noise detected was so loud, that there simply were not enough galaxies to produce them... they would have to be packed in much closer together. Even if all the galaxies in the universe only radiated in the radio frequencies, they still would not be able to produce the amount of radio noise observed. There is nothing that we know of in the universe that is energetic enough to produce the observed amount of radio noise. This

phenomenon is known as the space roar, and is an unsolved problem in astrophysics, cosmology and particle physics as well.

THE UNIVERSE AS A COMPUTER

There is a branch of physics and cosmology called digital physics, which attempts to explain the universe as an output of a computer program. This theoretical universal computer runs a program that figures out how the universe evolves, and our physical reality is the result of this program. Physicists have even attempted to figure out what kind of machine this universal computer could be, and some proposals include a Turing

machine where the cells change their state according to the codified rules. According to the cellular automata theory, the entire universe is a parallel computer. On the tiniest time scales, the universe is deterministic and extremely simple. The laws of physics emerge from local interactions. According to this model, there should be a limit to how fast information can travel, which neatly explains why photons have a fixed speed, which is the clock cycle of the universe. These cluster of theories predict that there can be alternate universe, to which photons are leaking from our universe. Photons coming from other universes would be virtual photons – which form dark matter. This is a



Jansky's Merry Go Round

Credit: NRAO

machine or a cellular automaton. A Turing machine is a relatively simple machine that modifies the symbols on a memory tape or strip, using a set of rules codified in a table. Despite the simplicity of the theoretical machine, a Turing machine can be created for any computer algorithm. A cellular automaton is a giant grid of cells, with a number of states (say on or off) and a set of rules that determine how a cell affects its neighbouring cells. Initially the entire grid is at a particular state of time, where $t=0$. With the advancement of time in increments of 1, there are generations of the entire grid,

form of matter that we cannot see, but the gravitational influence of which is required to explain why the galaxies have not flown away from each other yet. Over time, the amount of matter in the universe would reduce, and the amount of dark matter in the universe would increase. This would mean that in the early universe, there would be a lot more matter than we see in the universe today. This matter could be the source of the radio noise phenomenon we know as the space roar. This means that the space roar could be interpreted as evidence that the universe is in fact, one gigantic computer. **4**